Sahaj Bhadja

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Summary

Software Engineer pursuing an MSCS at USC with 2+ years building **low-latency backend systems**, **concurrent pipelines**, and **distributed services** in **C++**, Python, and Java, with strong **data structures**, **algorithms**, **multithreading**, and **OOP** foundations.

Education

University of Southern California, Los Angeles, CA — **August 2024 – May 2026**
Master of Science in Computer Science — *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India — **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering — *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife — **Los Angeles, CA**
Software Engineer — *September 2025 – Present*
- Built **distributed backend services** processing 5K+ monthly jobs with **fault-tolerant async orchestration**, explicit job-state transitions, and failure isolation, increasing throughput **3x** under concurrent load while sustaining 99%+ production reliability.
- Optimized **multi-tenant SQL access patterns**, indexing, and query execution for 500+ users, reducing request latency from **1.2s to 350ms** through algorithmic tuning of hot paths and data-access design.

Grad8 — **Los Angeles, CA**
Software Engineer Intern — *June 2025 – August 2025*
- Engineered a **concurrent batch-processing engine** using **multithreading** and ThreadPoolExecutor, processing 1K+ files per batch with fault-tolerant execution while handling **10K+ daily requests** under sustained peak load.
- Designed **production backend APIs** serving 150+ users with clean **OOP design** and CI/CD-driven releases, cutting feedback turnaround **60%** while maintaining stable response times during high-throughput operations.

USC AD Labs — **Los Angeles, CA**
ML Research Assistant — *January 2025 – May 2025*
- Built **large-scale data pipelines** over 100K+ trajectory records, applying efficient **algorithms and data structures** for reproducible, high-confidence benchmarking while improving signal quality for debugging model-driven decision systems.
- Architected a **transformer-based sequence model** with multi-head attention, improving long-horizon planning **18%** in simulation through rigorous evaluation, profiling, and performance-focused experimentation across closed-loop rollouts.

Aasma Technology Solutions — **India**
Software Developer Engineer - (SecureSync) — *August 2023 – July 2024*
- Built **object-oriented modules** using Singleton and Factory patterns for large-scale API integrations with AWS Glue and S3, improving response times **30%** across distributed, cross-platform production workloads.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito across 3 operating systems, increasing CI reliability **68%** and reducing regressions within a fast-paced, frequent release cycle.

Aasma Technology Solutions — **India**
Full Stack Developer Intern - (VibrantChem) — *May 2023 – August 2023*
- Developed **real-time WebSocket and microservice modules** syncing 2M+ records across eight workflows, improving throughput consistency and strengthening **low-latency responsiveness** for high-volume concurrent user-facing processes.
- Automated **Docker-based build and test pipelines** with Jenkins, cutting release time **50%**, improving deployment consistency, and reducing environment drift that previously slowed validation, troubleshooting, and handoffs.

Skills

Languages: **C++**, Python, Java, JavaScript/TypeScript, SQL, Bash
Core CS: Data Structures, Algorithms, Multithreading, Concurrency, OOP, System Design
Backend & Systems: Distributed Systems, REST APIs, FastAPI, Flask, Node.js, Spring Boot, Microservices
Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Query Optimization
Cloud & Delivery: AWS, Docker, Kubernetes, CI/CD, Jenkins, Git/GitHub, Linux
Practices: Unit/Integration Testing, Performance Optimization, Low-Latency Systems, Agile

Projects

CPU-Scheduling Simulator Live | *C++, Algorithms, OS Internals*
- Built a **C++ CPU-scheduling simulator** implementing FCFS, SJF, SRT, Round Robin, and multi-level queue **algorithms**, reducing average turnaround **25%** with Gantt visualization across 1K+ simulated process combinations.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered SQL workspace** across a **Java and TypeScript** codebase supporting 10+ databases, accelerating query authoring and schema exploration with Docker- and Maven-backed cross-platform development workflows.

FinNexus Live | *Next.js, Kubernetes, MongoDB, WebSockets*
- Engineered a **scalable fintech platform** handling 1M+ transactions for 50K+ users with **real-time updates**, token-based security, and Kubernetes-backed CI/CD, sustaining low-latency performance under high transaction volume.